

Executive Summary

Company Overview

Zepto is a quick-commerce business that offers a wide range of products across multiple categories, with a focus on product availability, competitive pricing, discounts, and efficient inventory management. The business maintains product-level data covering categories, product names, MRP, discount percentages, discounted selling prices, available quantities, product weights, stock status, and quantities.

As the business grows, management needs to understand **product value, pricing strategies, discount patterns, revenue potential, inventory levels, stock availability, and category performance** to make informed business decisions.

This SQL-based analysis addresses **18 key business questions**, connecting each question to a specific business problem and its potential business impact.

Business Problems

- 1. Top 10 Best-Value Products**
Identify products offering the highest discount percentages.
- 2. High-MRP Products Out of Stock**
Identify high-value products that are currently unavailable.
- 3. Estimated Revenue by Category**
Determine the estimated revenue potential across product categories.
- 4. High-MRP & Low-Discount Products**
Identify high-priced products with relatively low discounts.
- 5. Top 5 Categories by Average Discount**
Identify categories offering the highest average discounts.
- 6. Price per Gram for Products Above 100g**
Compare products based on price per gram to identify better-value options.
- 7. Inventory Segmentation — Low, Medium & Bulk**
Classify products based on inventory quantity levels.
- 8. Total Inventory Weight by Category**
Measure the total physical inventory weight across categories.
- 9. Highest Total Inventory Quantity by Category**
Identify categories holding the largest inventory quantities.
- 10. Top 10 Products by Estimated Revenue**
Identify products with the highest estimated revenue potential.

- 11. Categories with Highest Out-of-Stock Products**
Identify categories experiencing the highest number of stock-outs.
- 12. Overall Out-of-Stock Percentage**
Measure the proportion of products currently unavailable.
- 13. Products with Highest Discount Amount**
Identify products offering the largest absolute price reductions.
- 14. Average MRP & Selling Price by Category**
Compare average product pricing across categories.
- 15. Categories with Highest Potential Revenue**
Identify categories with the highest revenue potential from available inventory.
- 16. High Inventory & Low Discount Products**
Identify products with high inventory levels but relatively low discounts.
- 17. Inventory Quantity Distribution**
Determine the distribution of products across Low, Medium, and Bulk inventory segments.
- 18. Highest Average Price per Gram by Category**
Compare categories based on their average price per gram.

Business Questions

- 01.** What are the top 10 best-value products based on discount percentage?
- 02.** What are the products with high MRP but currently out of stock?
- 03.** What is the estimated revenue for each category?
- 04.** Which products have MRP greater than ₹500 and discount below 10%?
- 05.** What are the top 5 categories offering the highest average discount percentage?
- 06.** What is the price per gram for products above 100g, sorted by best value?
- 07.** How can products be grouped into Low, Medium, and Bulk inventory segments?
- 08.** What is the total inventory weight per category?
- 09.** Which product categories have the highest total inventory quantity?
- 10.** What are the top 10 products generating the highest estimated revenue?
- 11.** Which categories have the highest number of out-of-stock products?
- 12.** What percentage of products are currently out of stock?
- 13.** Which products have the highest discount amount?
- 14.** What is the average MRP and discounted selling price for each category?

15. Which categories have the highest potential revenue from available inventory?
16. Which products have high inventory quantity but relatively low discounts?
17. What is the distribution of products across different inventory quantity segments?
18. Which categories have the highest average price per gram?

Business Impact

1. **Top 10 Best-Value Products**
Helps optimize promotional strategies and highlight attractive customer offers.
2. **High-MRP Products Out of Stock**
Helps prioritize replenishment and minimize potential revenue loss.
3. **Estimated Revenue by Category**
Supports category prioritization and revenue-focused business planning.
4. **High-MRP & Low-Discount Products**
Helps evaluate pricing competitiveness and identify discount optimization opportunities.
5. **Top 5 Categories by Average Discount**
Helps evaluate discount intensity and improve promotional budget allocation.
6. **Price per Gram**
Supports value-for-money analysis and competitive pricing decisions.
7. **Inventory Segmentation**
Enables better stock monitoring and targeted inventory management.
8. **Total Inventory Weight**
Supports storage, handling, and logistics planning.
9. **Total Inventory Quantity**
Helps optimize stock allocation and inventory planning.
10. **Top 10 Products by Estimated Revenue**
Helps prioritize high-value products and focus sales efforts effectively.
11. **Out-of-Stock Products by Category**
Helps prioritize replenishment and reduce lost sales opportunities.
12. **Overall Out-of-Stock Percentage**
Provides a quick overview of inventory health and product availability.
13. **Highest Discount Amount**
Helps assess promotional effectiveness and optimize discount strategies.
14. **Average MRP & Selling Price**
Supports category-level pricing analysis and competitive positioning.

15. Potential Revenue by Category

Helps prioritize high-potential categories and improve inventory allocation.

16. High Inventory & Low Discount

Helps optimize pricing and reduce the risk of inventory buildup.

17. Inventory Quantity Distribution

Enables targeted inventory monitoring and more effective replenishment planning.

18. Average Price per Gram

Supports category-level pricing evaluation and identification of high-value product segments.

Tools & Technologies

- **Database:** Microsoft SQL Server
- **SQL Environment:** SQL Server Management Studio (SSMS)
- **Language:** SQL
- **Data Import:** BULK INSERT
- **Data Analysis:** Aggregations, filtering, grouping, conditional logic, subqueries, CTEs, and calculations
- **Data Validation:** NULL checks and record-count validation
- **Formatting:** CAST(), ROUND(), and DECIMAL formatting

The analysis uses SQL queries to answer the 18 defined business questions, applying techniques such as SELECT, WHERE, GROUP BY, ORDER BY, aggregate functions, TOP, CASE, subqueries, CTEs, and conditional calculations. This follows the technical framing and SQL-focused methodology demonstrated in the reference document.

SQL Analysis

The analysis focuses on:

- **Product Performance:** Identifying high-value and high-revenue products.
- **Pricing Analysis:** Comparing MRP, selling prices, discounts, and price per gram.
- **Discount Analysis:** Evaluating discount percentages and absolute discount amounts.
- **Inventory Management:** Assessing inventory quantities, weight, and product-level stock distribution.
- **Stock Availability:** Identifying out-of-stock products and categories.
- **Revenue Potential:** Estimating revenue at product and category levels.

- **Category Performance:** Comparing pricing, discounts, inventory, and revenue across categories.

Key Insights

The analysis is designed to uncover actionable insights across the following areas:

- **Product Value:** Identify products offering the highest discounts and best price-per-gram value.
- **Pricing Strategy:** Evaluate MRP, selling prices, discount percentages, and discount amounts.
- **Revenue Potential:** Identify products and categories with the highest estimated revenue potential.
- **Inventory Performance:** Analyze inventory quantities, weight, and product-level stock distribution.
- **Stock Availability:** Measure overall stock-out levels and identify categories with higher stock-outs.
- **Category Performance:** Compare categories based on revenue, inventory, pricing, discounts, and product value.
- **Inventory Efficiency:** Identify high-inventory products with relatively low discounts.
- **Inventory Segmentation:** Classify products into Low, Medium, and Bulk inventory segments.

Recommendations

Based on the business objectives addressed through the analysis, Zepto can consider the following actions:

1. **Prioritize replenishment** for high-MRP products and categories experiencing frequent stock-outs to minimize potential lost sales.
2. **Focus inventory investment** on high-revenue products and categories with strong revenue potential.
3. **Optimize discount strategies** by evaluating categories and products with exceptionally high discount percentages or discount amounts.
4. **Review high-inventory, low-discount products** and consider targeted promotions to improve inventory movement.
5. **Use price-per-gram analysis** to evaluate product value and support competitive pricing decisions.

6. **Optimize inventory allocation** by monitoring categories with high inventory quantities and physical inventory weight.
7. **Leverage inventory segmentation** to develop targeted replenishment and stock-monitoring strategies for Low, Medium, and Bulk products.
8. **Monitor stock-out percentage regularly** to maintain healthy product availability and improve customer experience.
9. **Prioritize high-potential categories** based on available inventory and estimated revenue to improve resource allocation.
10. **Continuously evaluate category-level pricing and discount patterns** to balance customer value, sales potential, and inventory efficiency.